

Oxygen-Vacancy Energetics and Dopant-Induced Holes in Pr- and Rb-Substituted Anatase TiO₂: A DFT+U Comparison

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Revision note (v4, 2 October 2026; delete before submission). This version implements the thirteen corrections of the Error Audit (E1–E13) and responds to the eighteen findings of the Reviewer/Editor Audit (R01–R18). All numerical values are unchanged from the FINAL draft and were re-derived from the Rydberg totals in the Supporting Information. Passages marked **[AUTHOR ACTION]** require information or calculations that only the authors can supply; they are listed again in the accompanying revision report.

Abstract

Oxygen vacancies (V_O) in wide-gap oxide electron-transport layers cause trap-assisted recombination; aliovalent substitution is the usual lever for controlling them, yet trivalent rare-earth and monovalent alkali dopants have rarely been compared under one protocol. We compare Pr^{3+} and Rb^+ substitution on the Ti site of anatase TiO_2 with spin-polarised PBE+U calculations ($U = 3.5$ eV on Ti-3d, $2 \times 2 \times 1$ supercell) and a common oxygen reference, $\mu_O = \frac{1}{2}E(\text{O}_2)$. The neutral V_O formation energy falls from +4.74 eV in pristine anatase to +1.29 eV in the Pr-substituted cell and -4.06 eV in the Rb-substituted cell; the 5.36 eV Pr-Rb contrast changes by 0.04 eV on a $2 \times 2 \times 2$ k-mesh. The ordering follows the formal electron count, one hole per Pr^{3+} and three per Rb^+ , so one vacancy over-compensates Pr and leaves one residual hole with Rb. That hole is an empty minority-spin O-2p state with $0.59 \mu_B$ on one oxygen bonded to Rb; the Rb ion itself moves $1.27 \text{ \AA}$ towards the vacancy, so relaxation also contributes to the Rb result. The Rb cells carry unpaired spin in every treatment tested; the +2-charged vacancy cell reaches a total magnetisation of $2.99 \mu_B$. With U on Ti-3d only the Pr cells converge to non-magnetic smeared solutions, but the vacancy-free Pr cell develops one unpaired spin when an O-2p

Hubbard term is added, so its spin state is method dependent. The results are consistent with charge-compensation control of V_O energetics; the two-dopant design does not separate oxidation state from size and bonding.

Keywords: DFT+U; anatase TiO_2 ; oxygen vacancy; aliovalent substitution; hole localisation; electron-transport layer.

1. Introduction

Indoor photovoltaics are a leading candidate to replace disposable batteries in the growing population of low-power sensors [1]. Under the 200–1000 lx illumination typical of indoor spaces the open-circuit voltage is limited by non-radiative recombination **[NEEDS CITATION: typical indoor illuminance and the low-light V_{oc} limitation; Pecunia et al. may serve if the full text supports both]**, and wide-gap absorbers such as $CsPbBr_3$ [2] are paired with n-type oxide electron-transport layers (ETLs), for example TiO_2 or SnO_2 [3]. The same oxides serve as electron-transport layers in organic light-emitting diodes [4]. In TiO_2 the oxygen vacancy (V_O) is the dominant native defect: microscopy, photoemission, transient spectroscopy and first-principles work agree that its two excess electrons can either delocalise into the conduction band or localise as Ti^{3+} centres, and that the localised configuration acts as a trap [5–14]. Controlling the equilibrium V_O population of an ETL is therefore a materials-design problem, and substitutional aliovalent doping is a widely used lever [15–18].

For trivalent dopants on the Ti^{4+} site the formal charge mismatch is -1 e per site. Several DFT+U and experimental studies of lanthanide-doped TiO_2 and related oxides describe this deficit as accommodated electronically, by rehybridisation at the valence-band edge or by a change of dopant oxidation state [19–23] **[AUTHOR ACTION: confirm from the full texts that each of these five studies describes electronic accommodation of the dopant charge; Ahmad et al. is an experimental photoelectrochemistry study of Gd-doped TiO_2 nanorods and Polliotto et al. concern zirconium titanate, so drop any that do not.]**. The picture is not universal: Iwaszuk and Nolan showed for trivalent dopants in bulk rutile that oxygen-vacancy compensation can compete with, and in some cases dominate, electronic compensation [17]. The coupling between a lanthanide dopant and an oxygen vacancy has been analysed for Ce-doped anatase [24] and Gd-doped CeO_2 [18], and Bahmanrokh et al. proposed a defect-equilibria formalism for Ce-doped anatase in which both channels appear [25] **[AUTHOR ACTION: confirm from the full text that both electronic and vacancy compensation appear in that formalism.]**. Pr in particular has been treated at the anatase (101) surface [26] and in the bulk [23], and is an attractive test case, here modelled as trivalent with a 4f-in-core dataset (Section 2.1), because its ionic radius (0.99 Å in six-fold coordination [27]) is far from that of Ti^{4+} (0.605 Å). For monovalent alkali dopants the mismatch is -3 e per site, and the available evidence points consistently to V_O formation as the compensation channel. Carey and Nolan found that Li, Na, K and Rb substitution in α - Cr_2O_3 strongly reduces the V_O formation energy, with the larger alkalis having the larger effect [16]. Hybrid-functional work on alkali-doped α - Bi_2O_3 reached a similar conclusion [28], and Tian et al. recently combined experiment and calculation to show that Rb doping, in synergy with lattice strain, promotes oxygen-vacancy formation in TiO_2 [15].

What is missing is a comparison of the two dopant families in the same anatase host with the same supercell, pseudopotentials, Hubbard treatment and oxygen reference; the Rb case of Tian et al., for example, is embedded in a strained photochromic system and is not paired with a trivalent dopant. Without such a comparison, differences between published trivalent and monovalent results cannot be separated from differences in host, protocol or reference state. The comparison is also a useful test of the Kröger-Vink bookkeeping that is routinely invoked for dopant selection [29]: a Pr^{3+} substitution should leave one hole and a Rb^+ substitution three, so a single neutral V_O (two electrons) should over-compensate Pr and under-compensate Rb.

This work provides that bounded comparison. Using one PBE+U protocol we compute the neutral V_O formation energy in pristine, Pr-substituted and Rb-substituted $2 \times 2 \times 1$ anatase supercells, characterise the spin state and electron count of each cell, and test the sensitivity of the localisation picture to an oxygen-site Hubbard term. We also report charged-cell spin trends, Bader-charge differences, a hybrid-functional check on the pristine cell, dispersion-corrected single points, a k-mesh verification and alternative vacancy sites. We state explicitly which conclusions the two-dopant design supports: it establishes the ordering and magnitude of the V_O energetics within this protocol and shows that they follow the formal charge count, but it does not isolate oxidation state from ionic radius or dopant-specific bonding, and it does not establish device-level recombination behaviour. The bulk formation energies reported here are the reference against which surface and interface V_O energetics in ETL films would be measured; interface models are outside the present scope. Section 2 describes the protocol, Section 3 reports structure, energetics, electronic structure and sensitivity tests, Section 4 discusses the results against prior work, Section 5 states the limitations and Section 6 concludes.

2. Computational Methods

2.1 Code, functional and pseudopotentials. All calculations were performed with Quantum ESPRESSO v7.5 [30,31] (hardware and wall times in Table S6) using the PBE exchange-correlation functional [32] and projector-augmented-wave (PAW) datasets from PSlibrary 1.0.0 [33]: Ti.pbe-spn-kjpaw_psl.1.0.0, O.pbe-n-kjpaw_psl.1.0.0, Pr.pbe-spdn-kjpaw_psl.1.0.0 and Rb.pbe-spn-kjpaw_psl.1.0.0. The valence configurations and electron counts of these datasets are listed in Table 1 because they fix the electron count of every supercell. The Pr dataset belongs to the PSlibrary “spdn” lanthanide family, which is generated for the trivalent ion with the 4f electrons frozen in the core; its 11 valence electrons comprise the 5s, 5p, 5d and 6s/6p shells (the generation configuration uses fractional 6s/6p occupations). The Pr-4f states are therefore not variational in the present calculations, and no conclusion about 4f occupation or 4f hybridisation can be drawn from them (Section 5). The Bader electron totals of the $\text{Pr}_{\text{Ti}} + \text{V}_\text{O}$ and $\text{Rb}_{\text{Ti}} + \text{V}_\text{O}$ cells (376.998 and 374.998 e, Table S3) reproduce the counts in Table 1 and confirm the valence assignment. Table S7 lists the MD5 checksums printed by pw.x and the atomic wavefunctions that projwfc.x reads from each dataset (Ti 3s/3p/3d, O 2s/2p, Pr 5p/5d, Rb 4s); the Pr dataset contains no 4f wavefunction. **[AUTHOR ACTION: add the MD5 checksum of the Pr dataset and the “Valence configuration” block of each UPF header to Table S7.]** Kinetic-energy cut-offs of 40 Ry (wavefunctions) and 320 Ry (charge density) were used throughout. No cut-off convergence test was performed; comparable anatase defect work uses 55 Ry [13], and the consequences are discussed in

Section 5. **[AUTHOR ACTION: quote the suggested minimum cut-offs printed in the headers of the four UPF files, and run one 55 Ry single point on Pr_perfect and Pr_VO to bound the cut-off error on E_f .]**

Table 1. Pseudopotential valence configurations and resulting supercell electron counts. N_e is the number of valence electrons per cell; all counts are for neutral cells.

“Formal carriers” is the Kröger-Vink bookkeeping for Pr^{3+} , Rb^+ , Ti^{4+} and O^{2-} with two electrons per neutral V_O ; it does not state where the carriers localise. The cell labels in the first column (e.g. Pr_VO for $\text{PrTi}_{15}\text{O}_{31}$) are used throughout the text.

Species / cell	Valence configuration (PSlibrary 1.0.0)	Z_{val}	Composition	N_e	Formal carriers
Ti (spn)	$3s^2 3p^6 3d^2 4s^2$	12	-	-	-
O (n)	$2s^2 2p^4$	6	-	-	-
Pr (spdn, 4f in core)	$5s^2 5p^6 5d^1 (6s, 6p)^2$	11	-	-	-
Rb (spn)	$4s^2 4p^6 5s^1$	9	-	-	-
Pristine_perfect	-	-	$\text{Ti}_{16}\text{O}_{32}$	384	none
Pristine_VO	-	-	$\text{Ti}_{16}\text{O}_{31}$	378	2 excess electrons
Pr_perfect	-	-	$\text{PrTi}_{15}\text{O}_{32}$	383	1 hole
Pr_VO	-	-	$\text{PrTi}_{15}\text{O}_{31}$	377	1 net excess electron
Rb_perfect	-	-	$\text{RbTi}_{15}\text{O}_{32}$	381	3 holes
Rb_VO	-	-	$\text{RbTi}_{15}\text{O}_{31}$	375	1 residual hole

2.2 Hubbard correction. A Hubbard term in the rotationally invariant Dudarev form [34], which is the form implemented behind the HUBBARD card of Quantum ESPRESSO 7.x, was applied to the Ti-3d manifold with $U_{\text{eff}} = 3.5$ eV using ortho-atomic projectors; U_{eff} is written U, U(Ti-3d) or U(O-2p) in the rest of the paper. Deák et al. note that this value is close to an ab initio determination of U for anatase [35], and it has been used for Er-doped anatase [20] and for anatase (101) defect chemistry [7]; other anatase work uses $U = 4.2$ eV [13], a value originally fitted for the rutile (110) surface [36], or larger values [14]. Literature U values obtained with other projector schemes are not strictly transferable to the ortho-atomic projectors used here, and the sensitivity of the present results to U is reported in Section 3.7. **[AUTHOR ACTION: verify against the full texts that Mazierski et al. and Yin et al. used $U(\text{Ti-3d}) = 3.5$ eV and that Deák et al. quote an ab initio anatase value near 3.5 eV.]** The DFT+U approach rests on the formulation of Anisimov et al. [37]; linear-response determination of U [38] was not performed for any cell in this work. For the oxygen-site tests of Section 3.4 an additional $U_{\text{eff}} = 5.5$ eV was placed on the O-2p manifold. This value is transferred from the Cr_2O_3 study of Carey and Nolan [16], where it was used together with a Cr-3d term to localise oxygen holes; it has not been derived for anatase and is treated here as a sensitivity parameter rather than a calibrated value.

2.3 Supercells. A $2 \times 2 \times 1$ anatase supercell with $a = b = 7.63632$ Å and $c = 9.77900$ Å (volume 570.25 Å³) was used, containing 48 atoms ($\text{Ti}_{16}\text{O}_{32}$) in the perfect cell and 47 after removal of one oxygen atom. Dopants were introduced by replacing one Ti atom with Pr or Rb, i.e. 6.25 % of cation sites; one V_O per cell corresponds to 3.1 % of oxygen

sites or $1.75 \times 10^{21} \text{ cm}^{-3}$. These concentrations are far above the dilute limit relevant to ETLs, and all absolute quantities below should be read as those of a concentrated periodic defect array. Lattice parameters were taken from a variable-cell relaxation of the pristine cell and held fixed for all defect cells. **[AUTHOR ACTION: state the k-mesh, cut-off and residual stress of that variable-cell relaxation.]** The production V_O is a first-shell oxygen of the dopant site: in the relaxed Rb_VO coordinates the vacant position is an equatorial oxygen site 1.95 Å from the ideal cation position (Section 3.1, Table S8), and the two doped cells have identical atom ordering in their outputs, consistent with the same vacancy. Three alternative sites in the Pr cell are examined in Section 3.7; a second first-shell candidate (index O3 of the orchestration script) was planned but not computed. **[AUTHOR ACTION: confirm from the Pr_VO input that the same oxygen was removed, and give the relaxed first-shell Pr-O distances in Table S8; the value ≈ 2.17 Å quoted for the production site in earlier drafts is well below the Shannon-radius sum of 2.39 Å for six-fold $\text{Pr}^{3+}\text{-O}^{2-}$.]**

2.4 Brillouin-zone sampling and electronic settings. Production relaxations used Γ -point sampling, a choice made in earlier work on larger (108-atom) anatase defect cells only for hybrid-functional calculations [14]. **[AUTHOR ACTION: confirm this against the published version of Arrigoni and Madsen; their arXiv preprint (1912.07260) uses denser meshes for DFT+U and a Γ -only grid only for the hybrid calculations.]** Single-point total energies on the Γ -relaxed geometries were recomputed on a $2 \times 2 \times 2$ Monkhorst-Pack mesh for all four doped cells, and the projected densities of states and Löwdin moments were obtained from non-self-consistent calculations on a $2 \times 2 \times 2$ mesh (Figure 4; Supporting Information, Section S2); the k-mesh dependence of the formation-energy contrast is reported in Section 3.7. All calculations were spin-polarised (nspin = 2). Fractional occupations were treated with Gaussian smearing. The single-point inputs derived from the production runs (U sweep, Section 3.7) use degauss = 0.005 Ry, whereas the converged pristine- V_O relaxation (Section S1) and the self-consistent runs that regenerated the densities for the projected densities of states used 0.01 Ry (Table S10). **[AUTHOR ACTION: confirm degauss = 0.005 Ry in the production relaxation inputs.]** With smearing, the total energy printed by pw.x is the smearing free energy; the entropy term is small when few states lie near the Fermi level but has not been tabulated. **[AUTHOR ACTION: report the “smearing contrib. (–TS)” line for every cell in Table S10; if it exceeds 0.01 Ry for any defect cell, report internal energies as well.]** The electronic self-consistency threshold was 1×10^{-8} Ry. Symmetry was handled by the code’s default detection. **[AUTHOR ACTION: state whether nosym or any magnetisation constraint (tot_magnetization, constrained_magnetization) was used in any run.]**

2.5 Magnetisation conventions and spin initialisation. Two quantities are reported throughout: the total (net) magnetisation $m = \int [\rho_{\uparrow}(r) - \rho_{\downarrow}(r)] dr$, which can be of either sign, and the absolute magnetisation $m_{\text{abs}} = \int |\rho_{\uparrow}(r) - \rho_{\downarrow}(r)| dr$. For a single unpaired electron, localised or delocalised, both are close to 1 μ_B , so neither quantity measures localisation; $m_{\text{abs}} > |m|$ indicates spin density of both signs. The run log records starting_magnetization = 0 on every species for the production relaxations, and the single-point inputs derived from them use the same setting; pw.x accepts explicitly set zero values with nspin = 2 and stops only when starting_magnetization is not set for any species. From a spin-symmetric start a moment develops only if the symmetric solution is unstable and numerical noise (here the randomised starting wavefunctions) breaks the symmetry, as happened for both Rb cells; the same start can leave a calculation in a non-

magnetic solution when a magnetic one exists. **[AUTHOR ACTION: confirm the starting magnetization values from the production input files.]** For this reason the Pr_VO cell was re-relaxed from a magnetic start with starting_magnetization = 0.3 on Ti and 0.5 on the dopant under a separate prefix, and the two total energies were compared (Section 3.3). These starting values are fractions of the valence charge in the Quantum ESPRESSO convention, not site moments in μ_B . The converged pristine- V_O relaxation started from 0.05 on Ti and O and ended non-magnetic (Section S1), and the self-consistent run preceding the Rb_VO projected densities of states started from 0.3 on Ti and reproduced the production Fermi energy (7.1329 eV). No magnetic start was tried for Pristine_perfect or, with U on Ti-3d only, for Pr_perfect. A wider search over site-specific and antiparallel seeds was not performed, and the non-magnetic solutions are therefore described below as the lowest-energy solutions among those tested rather than as established ground states.

2.6 Geometry optimisation. Ionic positions were relaxed with BFGS until the largest force component fell below 0.005 Ry/Bohr (0.129 eV/Å). For the doped cells, charge mixing was damped (mixing_beta = 0.1) and the BFGS history shortened (bfgs_ndim = 3) to suppress oscillation about shallow minima; the pristine V_O cell was first attempted with mixing_beta = 0.3 (mixing_mode = local-TF) and converged only after the changes described in Section S1. The final maximum forces were 0.0026, 0.0042, 0.0037 and 0.0040 Ry/Bohr (0.067, 0.108, 0.095 and 0.103 eV/Å) for Pr_perfect, Pr_VO, Rb_perfect and Rb_VO. The run log lists an earlier pristine perfect run as a single-point calculation with a residual force of 0.0225 Ry/Bohr (0.58 eV/Å); if the production pristine energy is also unrelaxed, relaxing it would lower E(Pristine_perfect) and raise the pristine formation energy of Table 2 by the relaxation energy. **[AUTHOR ACTION: state whether the production Pristine_perfect energy (−4275.01530 Ry) comes from a relaxed geometry and give its maximum residual force; if not, relax it and update Table 2.]** The Pr_VO relaxation was stopped manually after the force fell below the threshold while the BFGS step oscillated; the energy change over the last iterations was below 1×10^{-9} Ry. **[AUTHOR ACTION: state etot_conv_thr and forc_conv_thr of the production inputs; with the maximum force below 0.005 Ry/Bohr for several steps (Figure S3) and an energy change below 1×10^{-9} Ry, pw.x would normally have terminated by itself unless a tighter threshold was set.]** This tolerance is looser than the 0.01–0.03 eV/Å commonly used for defect energetics, and its consequences are discussed in Section 5. Force trajectories are shown in Figure S3.

2.7 Formation energies. The neutral V_O formation energy of host X was evaluated as

$$E_f(V_O^0; X) = E(X \text{ with } V_O, q = 0) - E(X \text{ perfect}, q = 0) + \mu_O, \text{ with } \mu_O = \frac{1}{2}E(O_2) + \Delta\mu_O \text{ and } \Delta\mu_O = 0,$$

where $E(O_2)$ is the total energy of an isolated O_2 molecule in a 12 Å cubic box computed with the same settings; it converged to the triplet ground state ($m = 2.00 \mu_B$) and $\frac{1}{2}E(O_2) = -41.51621$ Ry (−564.857 eV). This is the oxygen-rich electronic-energy reference; no finite-temperature gas-phase correction or competing-phase analysis was applied, and the values are conditional oxygen-removal energies from the already substituted cells rather than equilibrium vacancy concentrations [39]. The known overbinding of O_2 by PBE [14] shifts all absolute formation energies at this reference by a common amount and cancels in differences between hosts; no empirical correction was applied. Formation

energies were evaluated from unrounded Rydberg totals and are reported to 0.01 eV; the full-precision ledger is Table S9. The comparative quantity $\Delta E_f = E_f(V_O; \text{Pr}) - E_f(V_O; \text{Rb})$ is independent of the oxygen reference.

2.8 Additional calculations. (i) Charged cells: Pr_VO and Rb_VO were relaxed at $q = +1$ and $+2$ with a uniform compensating background; no image-charge or potential-alignment correction was applied, so these cells are used only for their spin state and not for transition levels. (ii) U sweep: single-point calculations on the Pristine_perfect, Pr_VO and Rb_VO geometries of the production runs at $U(\text{Ti-3d}) = 3.0, 4.0$ and 4.5 eV. (iii) HSE06: one single-point calculation on the production Pristine_perfect geometry with exact-exchange fraction 0.25 and screening parameter 0.106 Bohr^{-1} [40,41], Γ -only, 40 Ry, without the Hubbard term. **[AUTHOR ACTION: confirm the exact screening_parameter value in the input (0.106 Bohr^{-1} corresponds to $0.2 \text{ \AA}^{-1}$; the canonical HSE06 value is 0.11 Bohr^{-1}), the exx q-grid, ecutfock, the EXX convergence threshold and number of outer iterations, and that the +U term was switched off. The HSE06 total energy (-3926.67 Ry) differs from the PBE+U total of the same cell (-4275.02 Ry) by 348 Ry, which a change of functional alone cannot produce; state which pseudopotential datasets were used in the hybrid run and whether exact exchange was evaluated with the PAW datasets. If different datasets were used, the Fermi-energy comparison in Section 3.6 must be removed.]** (iv) Dispersion: single-point energies for Pr_VO and Rb_VO with the Grimme D3 correction and Becke-Johnson damping [42,43], without three-body terms, following the DFT+U+D3 protocol used for defective anatase by Torres et al. [44]; the single points were started from scratch on the production geometries. (v) Alternative vacancy sites: three additional Pr_VO cells with the vacancy at atom indices O1, O7 and O17 of the production input. (vi) Bader analysis of Pr_VO and Rb_VO from pp.x charge densities with the Henkelman grid algorithm [45,46]. **[AUTHOR ACTION: state the FFT grid used for the pp.x density, whether the PAW all-electron reconstruction was included, and the Bader code version (v1.05 is given in the SI).]** (vii) Projected densities of states, Löwdin charges and Löwdin spin polarisations from projwfc.x, on $2 \times 2 \times 2$ non-self-consistent runs (tetrahedron occupations; 260 bands for Pr_VO and 226 for Rb_VO) that start from a Γ -only self-consistent density regenerated on the final relaxation geometry; Gaussian broadening 0.01 Ry (0.136 eV), energy step 0.02 eV, spilling parameters 0.0030 (Pr_VO) and 0.0032 (Rb_VO) (Section S2). Löwdin populations depend on the projector basis and are used only to locate the spin density.

3. Results

3.1 Structural relaxation

All four doped supercells relaxed to maximum forces below the 0.005 Ry/Bohr threshold (Section 2.6; Pr_VO and Rb_VO trajectories in Figure S3). The pristine V_O cell, which required a modified mixing scheme to converge (Section S1), was stopped at 0.0056 Ry/Bohr , slightly above the threshold. Figure 1 shows the (010) atomic plane that contains the dopant site and the vacancy, at ideal positions and in the relaxed Rb_VO cell as read from its final coordinates (Table S8). The vacant site is an equatorial first-shell oxygen of the dopant site, $1.95 \text{ \AA}$ from the ideal cation position. In the relaxed cell the Rb ion has moved $1.27 \text{ \AA}$ from that position towards the vacancy and lies $0.87 \text{ \AA}$ from the vacant oxygen site; its four nearest oxygens are at 2.54 (two), 2.56 and $2.71 \text{ \AA}$ and the next two at 2.95 and $3.07 \text{ \AA}$. The two Ti atoms that were bonded to the removed oxygen have moved away from it, to 2.36 and $2.38 \text{ \AA}$ from the vacant site (ideal Ti-O distances 2.03 and $1.95 \text{ \AA}$). The large monovalent ion thus occupies part of the space vacated by

the oxygen, so structural relaxation is part of the energy released when the vacancy forms next to Rb (Sections 3.2 and 4.3). **[AUTHOR ACTION: the relaxed Pr_VO coordinates were not available for this revision; place the pw.x output of the relaxed Pr_VO cell in figures/data/ so that make_fig1_structures.py adds panel (c), and state the Pr displacement and Pr-O distances here.]** Because the residual forces of the doped cells (0.07–0.11 eV/Å) are larger than the 0.01–0.03 eV/Å typically used for defect energetics, bond lengths quoted here are indicative, and the sensitivity of the energetics to tighter relaxation has not been tested (Section 5).

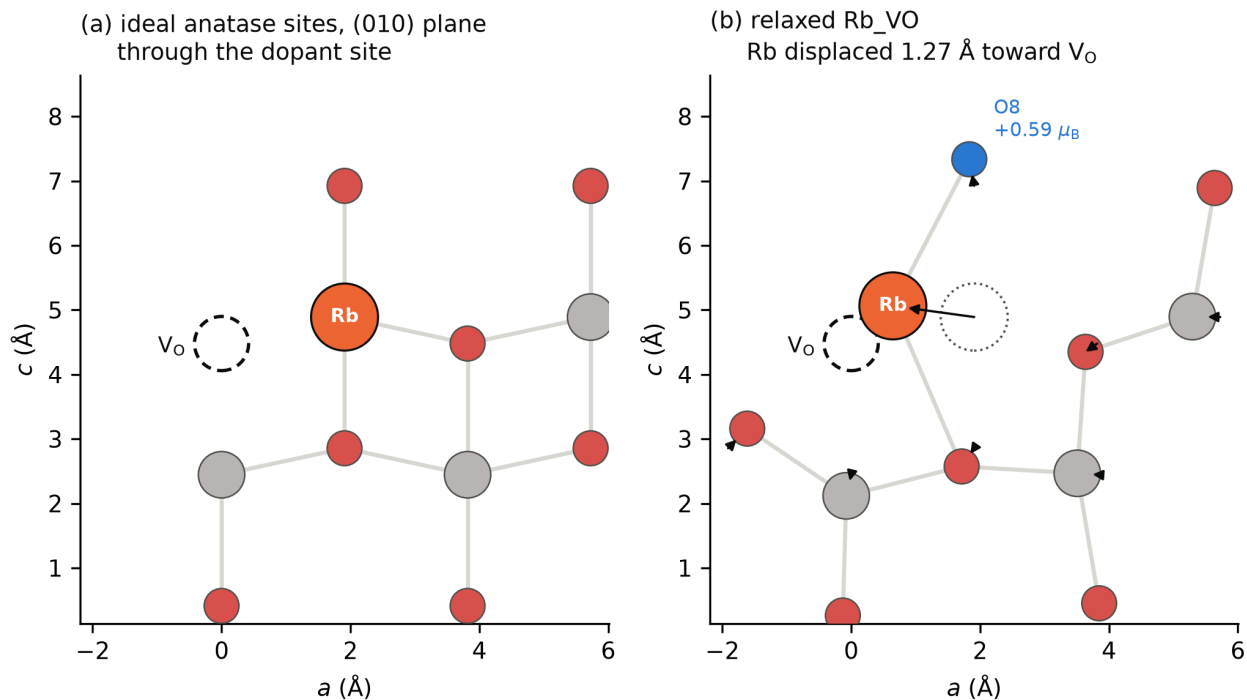


Figure 1. The (010) atomic plane containing the dopant site and the oxygen vacancy. (a) Ideal anatase positions built with the lattice parameters of this work ($a = 7.63632$ Å, $c = 9.77900$ Å, $u = 0.208$), with the dopant on the substituted Ti site and the vacant oxygen site marked by a dashed circle. (b) The same plane in the relaxed Rb_VO cell (PBE+U, $U(Ti-3d) = 3.5$ eV, Γ -only), from the final coordinates of the production relaxation. Arrows give the displacement of each in-plane atom from its ideal site after removal of the mean displacement; the dotted circle marks the ideal Rb site. The oxygen drawn in blue (O8) carries the largest Löwdin spin polarisation ($+0.59 \mu_B$; Section 3.6, Figure S1). Grey: Ti; red: O; orange: Rb. Only atoms in the plane are drawn, so each Ti shows fewer than six bonds. Plotted from the pw.x output by figures/make_fig1_structures.py.

3.2 Oxygen-vacancy formation energies

Table 2 lists the neutral V_O formation energies at the oxygen-rich reference $\mu_O = \frac{1}{2}E(O_2)$, and Figure 2 displays them. In pristine anatase $E_f = +4.74$ eV. In the Pr-substituted cell $E_f = +1.29$ eV, and in the Rb-substituted cell $E_f = -4.06$ eV. Both substitutions lower the cost of removing an oxygen atom near the dopant site relative to pristine TiO_2 , by 3.45 eV for Pr and 8.81 eV for Rb (differences of the unrounded values). At this reference the Pr cell remains uphill, so oxygen removal is not spontaneous; the Rb cell is downhill, so the vacancy-free Rb-substituted cell is unstable against oxygen loss. Part of the Rb reduction is structural: the Rb ion moves 1.27 Å into the space vacated by the oxygen (Section 3.1),

and the energy of this relaxation has not been separated from the electronic contribution. The Pr-Rb contrast ΔE_f is 5.36 eV (5.356 eV from the unrounded totals; the value 5.35 eV obtained by subtracting the rounded entries is a rounding artefact).

Table 2. Neutral oxygen-vacancy formation energies at $\mu_O = \frac{1}{2}E(O_2)$ (PBE+U, $U(Ti-3d) = 3.5$ eV, 40 Ry). Total energies are the converged Γ -only values in Ry; E_f is computed from the unrounded totals (full ledger in Table S9). The last column gives the single-point result on the Γ -relaxed geometry with a $2 \times 2 \times 2$ k-mesh. The Γ -to- $2 \times 2 \times 2$ shift is -0.10 eV (Pr) and -0.14 eV (Rb) and was not evaluated for the pristine formation energy; the Pr-Rb difference shifts by $+0.04$ eV. The $2 \times 2 \times 2$ Pr_VO and Rb_VO energies are runs of the k-mesh series (Table S2a), which start from scratch and therefore include small offsets from the relaxation endpoints (Section 3.7). Smearing, cut-off and relaxation-tolerance contributions to the absolute values have not been quantified (Sections 2.4 and 5).

Host X	E(X perfect) (Ry)	E(X + V _O) (Ry)	$\frac{1}{2}E(O_2)$ (Ry)	$E_f(V_O^0)$, Γ (eV)	$E_f(V_O^0)$, $2 \times 2 \times 2$ (eV)
Pristine TiO ₂	-4275.01530	-4233.15036	-41.51621	+4.74	- ^a
Pr _{Ti}	-4576.71224	-4535.10106	-41.51621	+1.29	+1.19
Rb _{Ti}	-4330.38890	-4289.17135	-41.51621	-4.06	-4.20
ΔE_f (Pr – Rb)				+5.36	+5.39

^a The pristine V_O cell was not computed at $2 \times 2 \times 2$ (the pristine perfect cell was, Section 3.7), so no $2 \times 2 \times 2$ formation energy can be evaluated for the pristine host. **[AUTHOR ACTION: compute E(Pristine_VO) at $2 \times 2 \times 2$ on the Γ geometry and complete this entry.]**

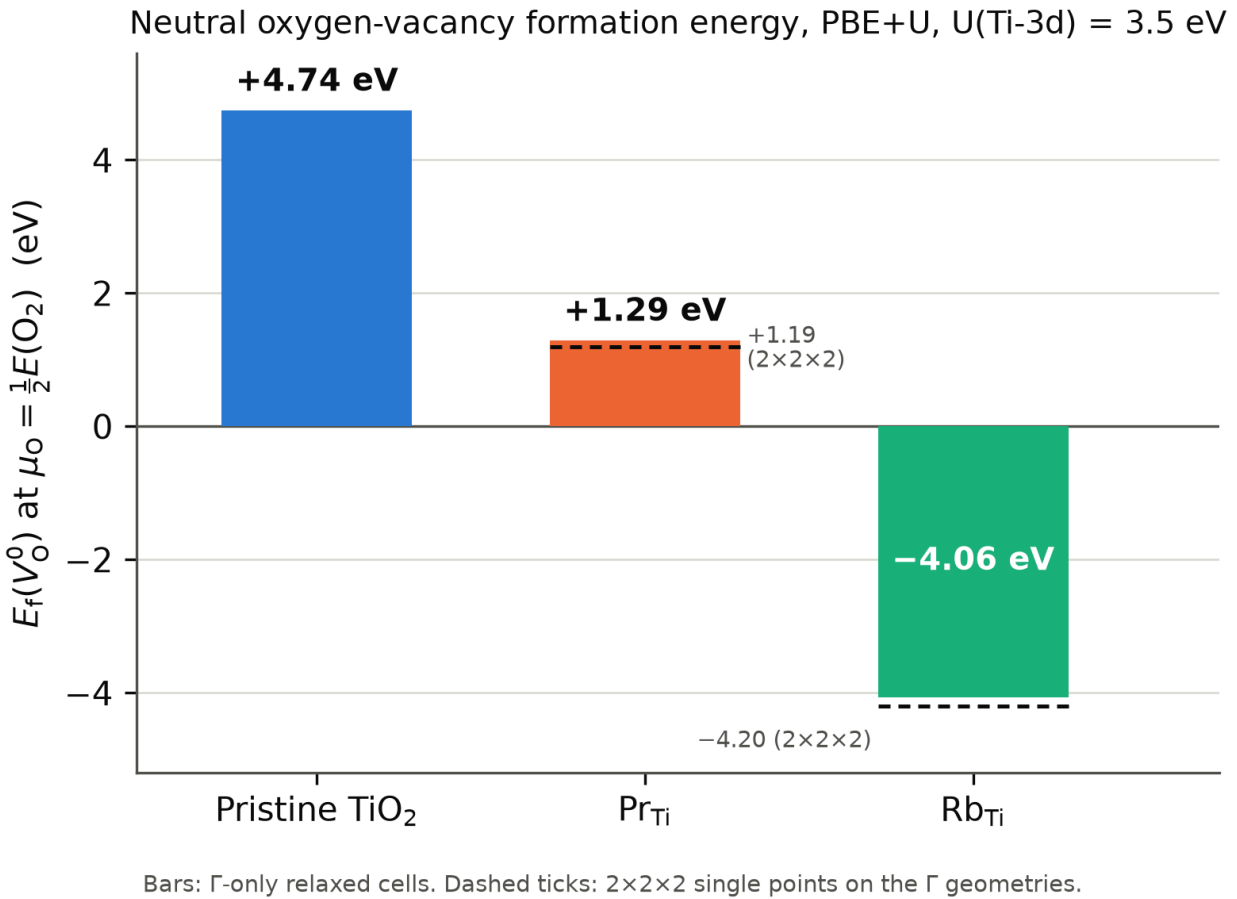


Figure 2. Neutral oxygen-vacancy formation energies at the oxygen-rich reference $\mu_O = \frac{1}{2}E(O_2)$ for pristine, Pr-substituted and Rb-substituted anatase (PBE+U, $U(\text{Ti-3d}) = 3.5 \text{ eV}$). Bars: Γ -only relaxed cells (this work; Table 2). Dashed ticks: single-point values at $2 \times 2 \times 2$ on the Γ geometries. Both substitutions lower the formation energy relative to pristine; the Pr cell stays positive and the Rb cell becomes negative.

The pristine value lies within the 4.0–5.0 eV range reported for the neutral bulk vacancy in anatase with DFT+U [14] and below the 5.3–5.5 eV obtained with HSE06 at the oxygen-rich limit [35,47] **[NEEDS CITATION: the bulk-anatase hybrid-functional V_O study of Deák et al. is Deák, Aradi and Frauenheim, Phys. Rev. B 92 (2015) 045204 (cited as such by Arrigoni and Madsen), rather than the surface paper cited here; if the value is taken from that study, add it to refs.py and cite it in place of Deak2014.]**, as expected for a semi-local functional with a PBE O_2 reference.

This is a consistency check, not a validation of the protocol. **[AUTHOR ACTION: verify the Arrigoni-Madsen, Boonchun and Deák values and their oxygen-chemical-potential conditions against the full texts. Note that Arrigoni and Madsen used a larger U (5.8 eV in their arXiv preprint), fitted elemental-energy corrections to their DFT+U energies and a reduced exact-exchange fraction (HSE15) in their hybrid calculations, so their values are not directly comparable with an uncorrected PBE+U value at a PBE O_2 reference.]** On the $2 \times 2 \times 2$ mesh the individual formation energies shift by -0.10 eV (Pr) and -0.14 eV (Rb) while the contrast moves from 5.36 to 5.39 eV, i.e. by $+0.04 \text{ eV}$ from the unrounded totals (defined as the $2 \times 2 \times 2$ contrast minus the Γ -only contrast). The comparative conclusion is therefore insensitive to Brillouin-zone sampling at the 0.05 eV level, although the absolute values carry a 0.10–0.14 eV mesh dependence. The direction of the Rb result (a reduction of E_f

relative to pristine) agrees with the combined experimental and computational finding of Tian et al. that Rb doping promotes V_O formation in TiO_2 [15]; that work supports the direction of the effect, not the specific supercell value.

3.3 Spin states and electron counts

Table 3 collects the total and absolute magnetisation and the Fermi energy printed by the code for each production cell, and Figure 3 plots the magnetisations. Table 1 shows that every doped cell has an odd number of electrons. A collinear calculation of an odd-electron cell can converge to $m = 0$ only through fractional occupations, so the non-magnetic solutions found for `Pr_perfect` and `Pr_VO` are smeared solutions in which the formal hole (`Pr_perfect`) or the formal net excess electron (`Pr_VO`) is carried by fractionally occupied states at E_F with equal spin-up and spin-down weight (conduction-band states in `Pr_VO`, Figure 4; not yet documented for `Pr_perfect`). They are not closed-shell singlets. The one magnetic trial performed for `Pr_VO` (Section 2.5) converged to $m = 1.00 \mu_B$, $m_{abs} = 1.13 \mu_B$. In smearing free energy (Section 2.4) it lies 0.281 eV above the non-magnetic energy logged with it for comparison and 0.394 eV above the final production energy (Table S9, runs 4, 24 and 25); this free energy includes a $-TS$ term that favours fractionally occupied solutions. The non-magnetic solution is therefore the lower of the two solutions tested at this smearing, and a broader search over spin seeds and local distortions would be needed to establish it as the ground state. For `Pr_VO` the smeared character is documented by the projected densities of states, in which the occupied conduction-band states have equal weight in the two spin channels, and by the Löwdin analysis, in which no site carries more than $0.001 \mu_B$ (Section 3.6). **[AUTHOR ACTION: document the occupations of the non-magnetic `Pr_perfect` solution in the same way.]** The isolated O_2 reference converged to its triplet ($m = 2.00 \mu_B$), which is the expected check of the spin-polarised pipeline.

The Rb cells behave differently. `Rb_perfect` converged to $m = 1.00 \mu_B$ ($m_{abs} = 1.00 \mu_B$) from the production starting magnetisation, and `Rb_VO` to $m = -0.97 \mu_B$ with $m_{abs} = 1.21 \mu_B$. Single-point calculations started from zero magnetisation on the final `Rb_VO` geometry, at the production U without and with the D3 term (Section 3.7) and at other U values (Table S1), converge instead to $m = 1.00 \mu_B$. The two solutions are nearly degenerate: the from-scratch Γ single point ($m = 1.00 \mu_B$, $E_F = 7.120$ eV) lies 6.4 meV above the last relaxation step ($m = -0.97 \mu_B$, $m_{abs} = 1.21 \mu_B$, $E_F = 7.133$ eV; Table S2a). The self-consistent run that preceded the projected densities of states, started from 0.3 on Ti, reproduced the production Fermi energy and is taken to be the production solution, with the opposite (arbitrary) sign of m . With three formal holes in `Rb_perfect` and one residual hole in `Rb_VO` (Table 1), the Ti-3d-only Hubbard treatment thus yields one unpaired spin in each Rb cell; the remaining two holes of `Rb_perfect` are spin-paired in this treatment ($m_{abs} = |m|$ excludes antiparallel moments), a point that the oxygen-site Hubbard test of Section 3.4 revisits. The excess of m_{abs} over $|m|$ in `Rb_VO` ($0.24 \mu_B$) indicates regions of spin density of the opposite sign, which can arise from spin polarisation of the surrounding ions whether the hole is localised or delocalised, so this quantity does not by itself fix the degree of localisation; the Löwdin analysis (Section 3.6) finds a weak antiparallel polarisation of the Ti sublattice ($-0.12 \mu_B$ in total), which accounts for part of the excess.

Table 3. Converged magnetisation and Fermi energy of the production cells ($U(\text{Ti-3d}) = 3.5 \text{ eV}$ only, Γ -only). m is the total magnetisation and m_{abs} the absolute magnetisation (Section 2.5). E_F is the Fermi energy printed by pw.x for each cell; the values are not aligned to a common reference and differences between chemically different cells are diagnostic only (Section 3.6).

Cell	N_e	m (μ_B)	m_{abs} (μ_B)	E_F (eV, unaligned)	Lowest solution among those tested
Pristine_perfect	384	0.00	0.00	7.89	non-magnetic
Pristine_VO	378	0.00 ^c	- ^c	[AUTHOR ACTION: E_F]	non-magnetic (Section S1)
Pr_perfect	383	0.00	0.00	7.40	non-magnetic (smeared)
Pr_VO	377	0.00	0.00	9.99	non-magnetic (smeared); magnetic trial +0.28 to +0.39 eV ^d
Rb_perfect	381	1.00	1.00	7.35	one unpaired spin
Rb_VO	375	-0.97	1.21	7.13	one unpaired spin (O-2p hole, Section 3.6)
O ₂ (12 Å box)	12	2.00	2.00	-	triplet

^c Reported as non-magnetic in Section S1; the printed values were not available for this revision. ^d Relative to the non-magnetic energy logged with the trial (+0.281 eV) and to the final production energy (+0.394 eV).

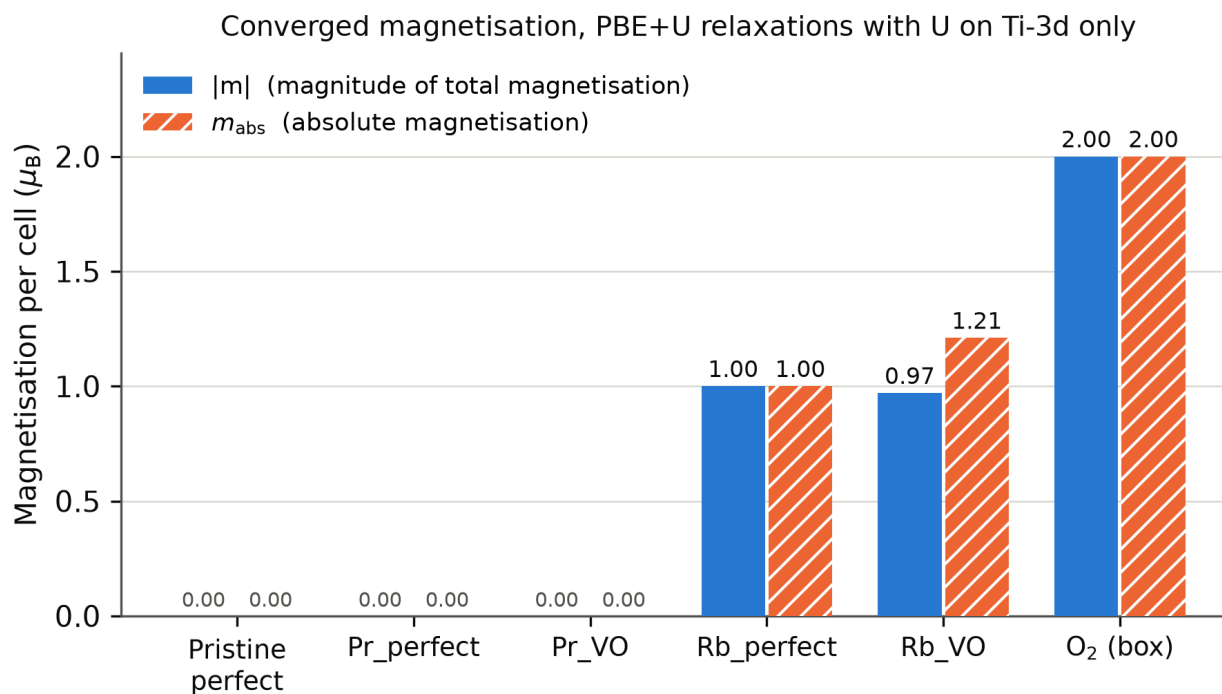


Figure 3. Magnitude of the total magnetisation ($|m|$) and absolute magnetisation (m_{abs}) of the production cells and of the O_2 reference (PBE+U relaxations with U on Ti-3d only). The Pr cells and the pristine cell converge to non-magnetic solutions; both Rb cells carry one unpaired spin, and Rb_VO shows $m_{abs} > |m|$.

3.4 Oxygen-site Hubbard correction: spin state of the dopant-induced holes

Because a Ti-3d-only Hubbard term provides no localising potential for holes on the oxygen sublattice, the three vacancy-free cells and Rb_VO were recomputed with an additional $U = 5.5$ eV on O-2p (Section 2.2), referred to below as the dual-U treatment. The pristine control is a single point; Pr_perfect and Rb_perfect were relaxed; the Rb_VO relaxation had not completed when its result was logged, so the Rb_VO entry is preliminary (Table S5). The magnetisations logged for these runs are total magnetisations m . **[AUTHOR ACTION: confirm from the output whether the dual-U Rb_VO relaxation finished, and give its final energy, m and m_{abs} .]** The results are collected in Table 4. The pristine control remains non-magnetic ($m = 0.00 \mu_B$), so the oxygen term does not induce magnetism in a cell without carriers; this control does not test whether the term biases hole-containing cells towards spin-polarised solutions. Rb_perfect changes from $m = 1.00$ to $m = 3.00 \mu_B$, i.e. the three formal holes of Table 1 become three parallel unpaired spins once the oxygen term is present (their localisation has not been established from site-projected moments); Rb_VO gives $m = 1.00 \mu_B$ (preliminary), the single residual hole after two have been filled by the vacancy electrons. Pr_perfect changes from the non-magnetic smeared solution to $m = 1.00 \mu_B$, with a maximum residual force of 0.0049 Ry/Bohr; the one formal hole of the Pr substitution thus becomes spin-polarised when the oxygen term is present. Its printed Fermi energy (6.69 eV) lies 0.53 eV below that of the dual-U pristine cell (7.22 eV), which is the direction expected when E_F is pinned at a hole state near the valence-band edge instead of lying in the gap of an insulating cell, whether or not that hole is localised, although, as for all Fermi energies in this work (Section 3.6), the unaligned difference is a qualitative indicator only. Because the localising term was placed on O-2p, the spins are expected to reside on oxygen, but this has not been confirmed from site-projected moments or spin densities. **[AUTHOR ACTION: report m_{abs} and the Löwdin site moments of dual-U Rb_perfect and Pr_perfect (projwfc.x on the existing outputs).]** The formal hole count of each dual-U cell is matched by its spin count: one per Pr^{3+} (a trivalent state imposed by the 4f-in-core dataset, Section 2.1), three per Rb^+ and one in Rb_VO, in line with the Kröger-Vink bookkeeping.

Table 4. Effect of an oxygen-site Hubbard term ($U(O-2p) = 5.5$ eV added to $U(Ti-3d) = 3.5$ eV) on the magnetisation of the vacancy-free and vacancy-containing cells. m : total magnetisation; m_{abs} : absolute magnetisation (not recorded for the dual-U runs).

Cell	m / m_{abs} , Ti-3d-only U (μ_B)	m , Ti-3d + O-2p U (μ_B)	Formal holes (Table 1)
Pristine_perfect	0.00 / 0.00	0.00 (single point)	0
Pr_perfect	0.00 / 0.00	1.00 (relaxed)	1
Pr_VO	0.00 / 0.00	not computed	-1 (net electron)
Rb_perfect	1.00 / 1.00	3.00 (relaxed)	3
Rb_VO	-0.97 / 1.21	1.00 (preliminary)	1

Two consequences follow. First, the dual-U spin count of the Rb cells equals the formal hole count and agrees with the $q = +2$ endpoint of the charged-cell series of Section 3.5, whereas the Ti-3d-only treatment gives one unpaired spin for the three holes of Rb_perfect. Second, the description of the Pr-substituted parent is method dependent: the hole is carried by a smeared non-magnetic solution in the Ti-3d-only treatment and by one unpaired spin, expected but not yet shown to reside on oxygen, in the dual-U treatment. The dual-U Pr_VO cell and the dual-U formation-energy cycles were not computed, so we do not know whether the formation energies of Table 2 change under the oxygen term; the statement that formation energies are less sensitive than localisation descriptors is an expectation, not a result of this work. **[AUTHOR ACTION: relax Pr_VO under the dual-U scheme (the other three dual-U cells exist), extract the dual-U total energies of Rb_perfect and Rb_VO from their outputs, and report $E_f(V_O)$ for both dopants with U(O-2p); this is the single most valuable addition before submission. A U(O-2p) sensitivity test (e.g. 4.5 and 6.5 eV on Pr_perfect) would show whether the one-hole localisation depends on the transferred value.]**

3.5 Charged-cell spin series

Relaxations of Pr_VO and Rb_VO at $q = +1$ and $+2$ (uniform background, no finite-size correction) converged in 216, 549, 293 and 360 ionic steps respectively; the raw total energies are listed in Table S9. Because the periodic-image electrostatics of a charged $2 \times 2 \times 1$ cell are uncorrected, these energies are not used to derive ionisation energies or transition levels. The spin state is used as a diagnostic. For Pr_VO both m and m_{abs} remain $0.00 \mu_B$ at $q = 0, +1$ and $+2$: the $q = +1$ cell (376 electrons) has no formal carriers, and the one formal hole of the $q = +2$ cell (375 electrons) remains unpolarised, as in Pr_perfect under the same treatment. For Rb_VO the magnitude of the total magnetisation rises from $0.97 \mu_B$ ($q = 0$) to $1.99 \mu_B$ ($q = +1$) and $2.99 \mu_B$ ($q = +2$); the absolute magnetisation is $1.21 \mu_B$ at $q = 0$ and $3.82 \mu_B$ at $q = +2$ (not recorded at $q = +1$), so the charged cells also contain spin density of the opposite sign. Each electron removed from the Rb cell therefore uncovers one further unpaired spin, and the $q = +2$ endpoint corresponds to the three holes per Rb^+ substitution found independently in the dual-U calculation of Section 3.4; the orbital character of the uncovered spins is not probed by this series. In the same Ti-3d-only treatment the neutral Rb_perfect cell, which has the same formal hole count (three) as Rb_VO at $q = +2$, carries only $1.00 \mu_B$ (Section 3.3), and the one-hole Pr_VO cell at $q = +2$ carries none, so the number of unpaired spins per hole is not a fixed property of this treatment and the series is a count of spins rather than a test of the hole count. **[AUTHOR ACTION: add m_{abs} of the $q = +1$ Rb_VO cell from its output.]** The two probes are not fully independent, since both are PBE+U calculations on the same geometry family, but they rely on different mechanisms (an added localising potential versus electron removal) and agree.

3.6 Electronic-structure diagnostics

Fermi energies. The Fermi energies printed for the production cells (Table 3) are 7.89 eV (pristine), 9.99 eV (Pr_VO) and 7.13 eV (Rb_VO). They are referenced to each cell's own average electrostatic potential and are not aligned to a common vacuum or core level; in the insulating pristine cell the printed value lies in the gap at a position set by the smearing rather than by a physical level (the same Γ -only cell prints 7.89 eV in the production run and 8.24 eV in a single point of the k -mesh series at the same total energy, Table S2a). The differences ($+2.10$ eV for Pr_VO and -0.76 eV for Rb_VO relative to pristine; 2.86 eV between the two doped vacancy cells) should therefore be read as

qualitative indicators and not as band offsets or carrier-density shifts. With that caveat, the large upward shift in Pr_VO is consistent with the net excess electron of Table 1 occupying conduction-band states in a cell containing one carrier per $570 \text{ \AA}^3$ ($1.75 \times 10^{21} \text{ cm}^{-3}$), and the small downward shift in Rb_VO is consistent with a partly filled oxygen-derived state near the valence-band edge. At realistic ETL carrier densities (10^{17} – 10^{19} cm^{-3}) the corresponding conduction-band filling would be far smaller; no estimate is attempted here. **[AUTHOR ACTION: if the Fermi-energy comparison is retained, align the cells by the average O-1s or Ti-2p core level, or by the macroscopic average potential far from the defect, and report the valence-band maximum and conduction-band minimum of each cell.]**

Projected densities of states and site moments. Figure 4 shows the spin-resolved projected densities of states of Pr_VO and Rb_VO from $2 \times 2 \times 2$ non-self-consistent calculations on the relaxed cells (Section 2.8), and Figure S2 the spin-summed densities of states. In Pr_VO the O-2p valence band ends about 2.9 eV below E_F and the Fermi level lies in the Ti-3d conduction band. The occupied conduction-band states within 0.6 eV below E_F have equal weight in the two spin channels (0.70 states each in the broadened DOS, 95 % Ti-3d), no feature appears inside the gap at this broadening, and the Pr-5d projection is small at both band edges. No site carries a Löwdin spin polarisation above $0.001 \mu_B$, so the net excess electron is spin-paired over delocalised conduction-band states: this is the smeared solution of Section 3.3. The Pr-4f states are frozen in the core and cannot appear (Section 2.1). In Rb_VO the Fermi level lies at the top of the O-2p valence band and an empty state lies just above it in one spin channel only: between E_F and $E_F + 0.6 \text{ eV}$ the minority channel holds 0.79 states and the majority channel none, and 93 % of the minority weight is O-2p (Ti-3d 5 %). This empty minority-spin state is the residual hole of Table 1. The Löwdin moments locate it (Figure S1, Section S2.2). Of the total of $+1.01 \mu_B$, $+0.59 \mu_B$ sits on a single oxygen, O8, and $+0.11 \mu_B$ on each of two further oxygens, O6 and O17; the oxygen sublattice carries $+1.12 \mu_B$ in total, the Ti sublattice $-0.12 \mu_B$ (at most $-0.03 \mu_B$ on any Ti) and Rb none. O8 is bonded to the displaced Rb ion ($2.56 \text{ \AA}$) and to two Ti atoms ($1.96 \text{ \AA}$) and lies $3.40 \text{ \AA}$ from the vacant site; it is also the oxygen with the largest Bader depletion (Section 3.7). The residual hole is therefore an O-2p hole centred on one oxygen next to the dopant, with smaller weight on its oxygen neighbours and no Ti^{3+} character. The sum of the magnitudes of the Löwdin moments ($1.24 \mu_B$) is close to m_{abs} ($1.21 \mu_B$). Löwdin moments depend on the projector basis, so these numbers locate the spin density rather than quantify it. The separation between the valence-band top and the conduction-band onset is about 2.4–2.5 eV in both cells (onsets taken where the total DOS reaches $0.5 \text{ states eV}^{-1}$). In the run used for Figure 4 the net moment is positive, so the hole appears in the spin-down channel; the sign of m is arbitrary in a collinear calculation, and the production relaxation ($m = -0.97 \mu_B$) corresponds to the same solution with the spins interchanged (Section 3.3). The Fermi energy of Pr_VO in Figure 4 (10.31 eV, $2 \times 2 \times 2$ non-self-consistent) differs from the Γ -only self-consistent value (9.99 eV) because of the k-mesh dependence of the conduction-band filling; the Rb_VO Fermi energy is mesh-stable (7.13 versus 7.14 eV). **[AUTHOR ACTION (optional): add the Rb_VO spin-density isosurface (pp.x, plot_num = 6) as Figure S4 to complement the Löwdin moments.]**

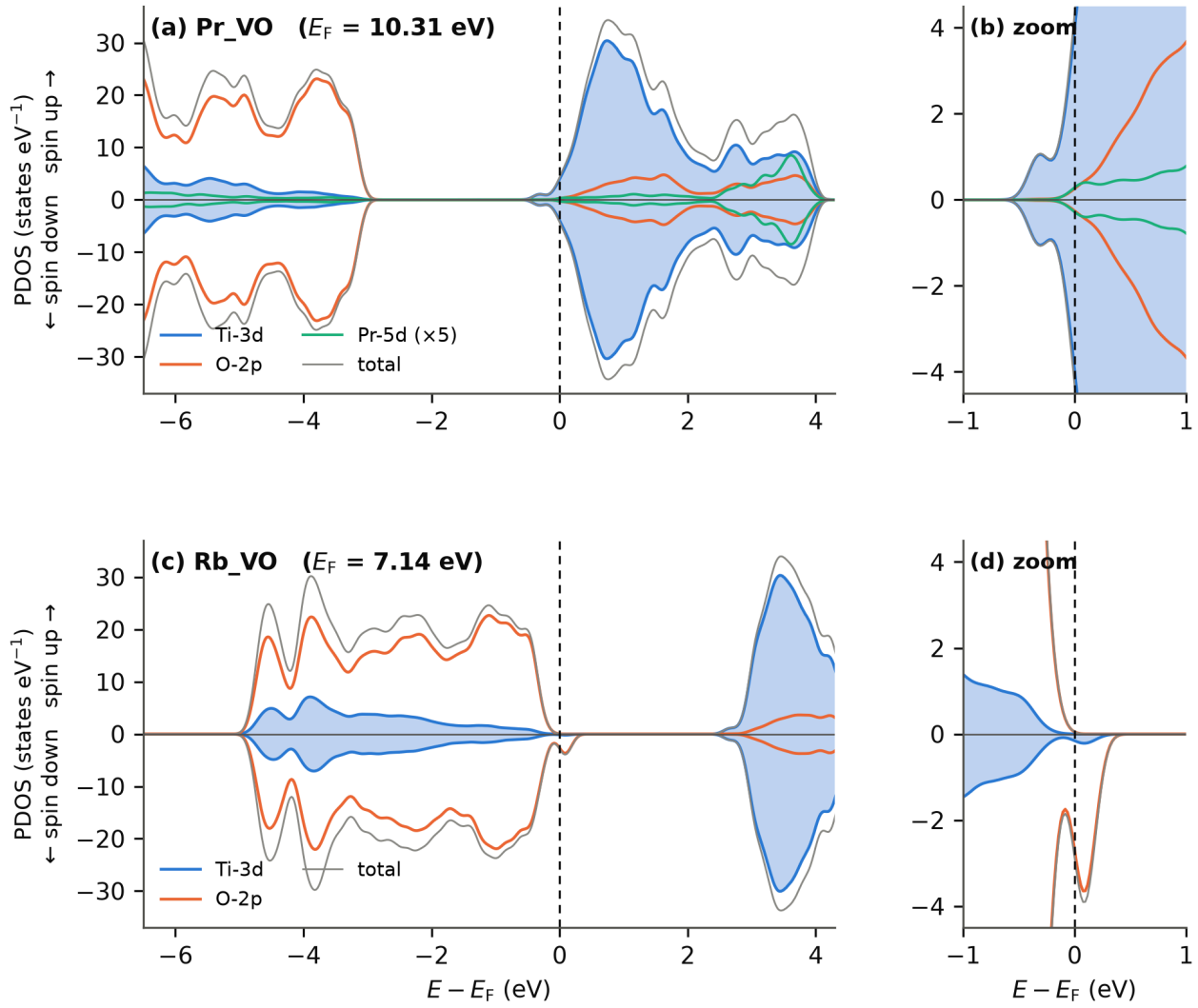


Figure 4. Spin-resolved projected densities of states of (a) Pr_VO and (c) Rb_VO, with (b, d) the region within ± 1 eV of E_F , from $2 \times 2 \times 2$ non-self-consistent calculations on the relaxed cells (PBE+U, $U(\text{Ti-3d}) = 3.5$ eV; tetrahedron occupations; Gaussian broadening 0.01 Ry). Spin up is plotted upwards and spin down downwards; energies are relative to each cell's Fermi level (10.31 and 7.14 eV). Filled blue: Ti-3d summed over 15 Ti; orange: O-2p summed over 31 O; green: Pr-5d multiplied by 5; grey: total. The Rb dataset has no d projector, so no dopant-d channel exists for Rb_VO. In (b) the occupied conduction-band states have equal weight in the two spin channels (0.70 states each between -0.6 eV and E_F , 95 % Ti-3d). In (d) the empty state just above E_F appears in the spin-down channel only (0.79 states between E_F and $+0.6$ eV, 93 % O-2p; spin up 0.00). Plotted without smoothing from the summed projwfc.x output by figures/make_pdos_figures.py.

Hybrid-functional check on the pristine cell. A single-point HSE06 calculation on the production Pristine_perfect geometry (Section 2.8) gave a non-magnetic solution ($m_{\text{abs}} = 0.00 \mu_B$), a total energy of -3926.66858 Ry and a Fermi energy of 8.286 eV, 0.40 eV above the PBE+U value. The shift of the printed Fermi energy is not a band-gap measurement, and the hybrid total energy is not comparable with the PBE+U totals (Section 2.8). The check establishes only that the non-magnetic pristine state survives the change of functional, which is expected for a closed-shell cell without carriers; no hybrid calculation was performed on the doped or defective cells, so the dopant contrast of Table 2 and the localisation results of Sections 3.3–3.5 have not been verified at the hybrid

level. **[AUTHOR ACTION: if the HSE06 result is retained, report the PBE+U and HSE06 band gaps of the pristine cell from the eigenvalues and resolve the pseudopotential question raised in Section 2.8.]**

3.7 Sensitivity tests

Hubbard U. Single-point calculations at $U(\text{Ti-3d}) = 3.0, 4.0$ and 4.5 eV on the production Pristine_perfect, Pr_VO and Rb_VO geometries (Table S1) leave the magnetisations unchanged in kind. Pr_VO is non-magnetic at every U , and Rb_VO carries one unpaired spin at every U ($m = 1.00 \mu_B$ in the from-scratch single points, versus $m = -0.97 \mu_B$, $m_{\text{abs}} = 1.21 \mu_B$ in the final relaxation step at $U = 3.5$ eV; see Section 3.3 for the two solutions). The printed Fermi energies of Pr_VO and Rb_VO vary by at most 0.26 eV across the 1.5 eV range; that of the insulating pristine cell lies in the gap and is not a physical level (Section 3.6). The total energies vary smoothly with U , by 6.1–6.6 eV per 0.5 eV step in all three cells. The qualitative Pr/Rb distinction in spin state is therefore not tied to the particular $U(\text{Ti-3d})$ used at fixed geometry; formation energies were not recomputed in the sweep.

k-mesh. For Pristine_perfect, Pr_VO and Rb_VO the total energy was computed at Γ , $2 \times 2 \times 1$, $2 \times 2 \times 2$ and $3 \times 3 \times 2$ meshes at $U = 3.5$ eV. The Γ -only energies differ from the $3 \times 3 \times 2$ values by 1.05, 1.08 and 1.16 eV per cell (22, 23 and 25 meV per atom), whereas the $2 \times 2 \times 1$ energies are within 1.15, 1.05 and 0.42 meV per atom of $3 \times 3 \times 2$. The Γ -only energies lie below the $3 \times 3 \times 2$ values, and the $2 \times 2 \times 2$ energies of the same series lie 0.2, 1.5 and 2.1 meV per atom above them for Pr_VO, Rb_VO and Pristine_perfect (Table S2a), so convergence beyond $2 \times 2 \times 1$ is at the 1–2 meV per atom level and is not monotonic across mesh families. The Γ -only production energies are therefore not converged in absolute terms, and the formation energies of Table 2 rely on cancellation between the perfect and defective cell of the same host rather than on convergence of each total energy. The $2 \times 2 \times 2$ column of Table 2 quantifies the residual: -0.10 eV (Pr), -0.14 eV (Rb) and $+0.04$ eV on their difference. Because the k-mesh single points start from scratch, the Γ -to- $2 \times 2 \times 2$ shifts include the differences between the from-scratch and relaxation-endpoint solutions at Γ (0.038 eV for Pr_VO and 6.4 meV for Rb_VO, Table S2a), which lie within the stated mesh dependence. A complete formation-energy cycle exists only at Γ and, for the doped hosts, at $2 \times 2 \times 2$. The total magnetisations are mesh-independent: $0.00 \mu_B$ for Pristine_perfect and Pr_VO and 0.99 – $1.00 \mu_B$ for Rb_VO at every mesh, the latter being the from-scratch Rb_VO solution of Section 3.3.

Dispersion. D3(BJ) single points on the relaxed Pr_VO and Rb_VO cells gave dispersion contributions of -10.29 and -10.09 eV (Table S5). Because D3 is an additive geometric correction, the electronic part of each single point should reproduce the production total energy. It does so to 0.006 eV for Rb_VO but lies 0.038 eV above the production value for Pr_VO. The electronic part of the Pr_VO D3 single point is identical, to 10^{-5} Ry, to the from-scratch Γ single point of the k-mesh series (Table S2a), so the offset belongs to the from-scratch SCF solution of the smeared Pr_VO cell and not to the D3 term. The dispersion energies of the perfect substituted cells and of the O_2 reference were not computed, so the dispersion correction to $E_f(\text{VO})$ and to ΔE_f cannot be reconstructed from these results: the correction to the contrast requires $[\text{D3}(\text{Pr_VO}) - \text{D3}(\text{Pr_perfect})] - [\text{D3}(\text{Rb_VO}) - \text{D3}(\text{Rb_perfect})]$, of which only the two defective-cell terms are available. The effect of dispersion on the vacancy energetics is therefore undetermined. The D3 single points give $m = 0.00 \mu_B$ (Pr_VO) and $1.00 \mu_B$ (Rb_VO) and Fermi energies within 0.02 eV of the production values; the Rb_VO value is the from-scratch solution discussed

in Section 3.3, not the production solution ($m = -0.97 \mu_B$, $m_{\text{abs}} = 1.21 \mu_B$). **[AUTHOR ACTION: run D3 single points on Pr_perfect, Rb_perfect and the O₂ reference (about 6 h per cell on the hardware listed in Table S6) so that both the individual formation energies and their difference can be corrected, or delete this paragraph.]**

Alternative vacancy sites in the Pr cell. Three further Pr_VO cells were relaxed with the vacancy at the oxygen indices O1, O7 and O17 of the production input (Table S4). Sites O1 and O17 are symmetry-equivalent second-shell positions (Pr-O = 3.99 Å) and give $E_f = +2.07$ eV each, agreeing to 0.1 meV; the more distant site O7 (4.94 Å) gives +2.72 eV. The production site is thus the lowest of the four sampled sites by 0.78 eV. This is a comparison among selected configurations in a $2 \times 2 \times 1$ cell, where “distant” sites are still within one lattice parameter of the periodic images of the dopant. It indicates a preference for the production site over the two more distant positions sampled but is not a global search. The production site is itself a first-shell oxygen of the dopant site (Section 2.3); a second first-shell candidate was not computed, site multiplicities are not given, and no corresponding sampling was performed for the Rb cell. The positions of the removed oxygens are listed in Table S4. **[AUTHOR ACTION: if possible, add two Rb_VO sites (a second first-shell and a second-shell oxygen).]**

Bader charges. Bader analysis of the relaxed Pr_VO and Rb_VO cells (Table S3) gives dopant charges of +2.14 e (Pr) and +0.82 e (Rb) relative to the PAW valence, consistent with the +3 and +1 formal states after the usual reduction of Bader charges relative to formal values. **[AUTHOR ACTION: these two values are inconsistent with the mean Ti and O populations of Table S3, from which the sum rule gives +2.04 e (Pr) and +0.87 e (Rb); re-extract all four numbers from ACF.dat. Also state that the two cells share the same atom ordering and which oxygen index was removed in each, and report the grid used; Bader grid convergence was not tested.]** Relative to the Pr_VO cell, the 31 oxygen atoms of Rb_VO hold 0.97 e less in total, with the depletion concentrated on one oxygen bonded to the displaced Rb ion (−0.26 e, O8; Rb-O8 = 2.56 Å, 3.40 Å from the vacant site) and a second oxygen (−0.14 e, O30; 3.07 Å from Rb), five further oxygens contributing −0.05 to −0.09 e each. The 15 Ti atoms show a uniform mean depletion of −0.012 e (−0.18 e in total) and no Ti differs by more than 0.05 e. No Ti site shows the excess of a few tenths of an electron, relative to the other Ti sites, that a localised Ti^{3+} centre would produce, although the expected size of that signal was not calibrated on a reference cell containing a Ti^{3+} centre, so the test is indicative rather than decisive. The comparison is between cells of different composition, so the differences contain chemical polarisation by the dopant as well as carrier redistribution, and Bader charges are not orbital occupations; the two cells also differ by two carriers (the net excess electron of Pr_VO and the residual hole of Rb_VO, Table 1), whose contributions to the oxygen depletion this comparison cannot separate; the result is therefore compatible with an oxygen-centred hole in Rb_VO, but it is not a hole count and carries no spin information. The Löwdin analysis of Section 3.6 supplies the spin information: the oxygen with the largest Bader depletion, O8, also carries the largest spin moment (+0.59 μ_B). A same-composition comparison (Rb_VO at $q = 0$ versus $q = +1$, or the spin density itself) would be more direct.

4. Discussion

4.1 The energetic ordering follows the formal electron count. The three formation energies of Table 2 fall in the order pristine (+4.74 eV) > Pr (+1.29 eV) > Rb (−4.06 eV), which is the order of the formal hole count of the vacancy-free cells (0, 1, 3; Table 1) and also of increasing cation radius (Section 4.3). A neutral vacancy releases two electrons. In the pristine cell both must occupy conduction-band-derived states, and the full cost of oxygen removal is paid. In the Pr cell one of the two can instead fill the single acceptor-like hole introduced by Pr^{3+} , and in the Rb cell both can fill two of the three holes introduced by Rb^+ . Transferring an electron from a conduction-band-like state to an oxygen-2p-derived hole near the valence-band edge recovers an energy of the order of the band gap, so reductions of a few eV per transferred electron are expected; the reductions found (3.45 eV for one electron in the Pr cell, 8.81 eV for two in the Rb cell) are of that order but not equal per electron. The valence-to-conduction-band separation in the projected densities of states of the two vacancy cells is about 2.4–2.5 eV (Section 3.6), smaller than either per-electron reduction (3.45 eV for Pr, 4.40 eV per electron for Rb), so electron transfer across the gap does not account for the reductions on its own. The remainder must come from the electrostatic interaction between the vacancy and the dopant, which the dependence of E_f on the vacancy site in the Pr cell indicates (Section 3.7), and from lattice relaxation, which in the Rb cell includes a 1.27 Å displacement of the Rb ion into the space vacated by the oxygen (Section 3.1); the two-dopant design cannot decompose the formation-energy difference into these contributions.

4.2 Relation to previous work. The Rb result parallels, in anatase, the behaviour that Carey and Nolan found for alkali dopants in $\alpha\text{-Cr}_2\text{O}_3$, where the V_O formation energy is strongly reduced and oxygen holes are localised with an O-2p Hubbard term [16]. It also agrees in sign with the experimental vacancy-formation trend reported for Rb-doped TiO_2 by Tian et al. [15]. Because only one alkali was studied here, the radius dependence found by Carey and Nolan across Li–Rb is not tested. The Pr result is consistent with the trivalent-dopant picture of Iwazuk and Nolan for rutile [17]: the vacancy is not spontaneous at the oxygen-rich reference, but its cost is reduced by 3.45 eV relative to the pristine host, so vacancy formation near Pr becomes accessible as the oxygen chemical potential is lowered; one vacancy per Pr over-compensates the dopant (Table 1), and the fully compensated $2\text{Pr}_{\text{Ti}} + V_O$ configuration that corresponds to vacancy compensation of a trivalent dopant was not computed. With $E_f(V_O; \text{Pr}) = 1.29 \text{ eV} + \Delta\mu_O$, oxygen removal near Pr has a negative formation energy once μ_O falls more than 1.29 eV below $\frac{1}{2}E(\text{O}_2)$, a threshold that carries the common offset of the PBE O_2 reference (Section 2.7); whether that lies within the stability field of TiO_2 against lower oxides was not evaluated (Section 5). The preference for the production vacancy site over the two more distant sites sampled (Section 3.7) is in the same direction as the Pr- V_O association reported for Pr adsorbed at the anatase (101) surface by Vural et al. [26], although adsorption and bulk substitution are different situations and only four sites were sampled here. **[AUTHOR ACTION: confirm from the full text of Vural et al., with a page or figure reference, that they report a Pr- V_O association; the institutional records of their abstract describe Pr-induced gap states, optical activity and benzoic-acid adsorption but do not mention oxygen vacancies. If the association is not reported there, delete this sentence.]** The spin polarisation of the Pr-induced hole when an oxygen-site U is present parallels the oxygen-hole states reported for trivalent dopants in rutile by Iwazuk and Nolan [17].

4.3 What the two-dopant comparison does and does not isolate. Pr^{3+} and Rb^{+} differ in formal charge (-1 versus -3 e relative to Ti^{4+}), in ionic radius (0.99 versus 1.52 Å [27]), in bonding (a 5d-bearing trivalent cation versus a closed-shell alkali) and in accessible oxidation states. Both are larger than Ti^{4+} (0.605 Å), so the comparison does show that a large size mismatch is compatible with either sign of $E_f(\text{V}_\text{O})$ at the oxygen-rich reference; it does not hold size constant, and no strain or fixed-geometry decomposition was computed, so the contrast cannot be apportioned between electronic and elastic contributions. What can be said is that the sign of the contrast and the size of the reductions are consistent with the electron count; because the ordering $\text{pristine} > \text{Pr} > \text{Rb}$ is also the order of increasing cation radius (0.605, 0.99 and 1.52 Å) and Carey and Nolan found a radius dependence at fixed alkali valence [16], a size-based explanation of the ordering is not excluded. The relaxed Rb_VO cell gives a direct indication of the elastic part: the Rb ion moves 1.27 Å off the cation site into the space vacated by the adjacent oxygen (Section 3.1), a relaxation channel that only an adjacent vacancy opens and that a smaller ion would exploit less, but whose energy has not been separated from the electronic contribution. Isolating the two variables would require a controlled series within the same protocol, for example La^{3+} , Y^{3+} and Sc^{3+} (same valence, radii from 1.03 to 0.745 Å [27]) and K^{+} and Na^{+} (same valence as Rb^{+} , smaller radii), together with fixed-geometry single points to separate electronic from relaxation contributions.

4.4 The Pr-substituted parent is method dependent. The vacancy-free Pr cell is non-magnetic with U on Ti-3d only and carries one unpaired spin ($m = 1.00 \mu_B$), expected but not yet shown to reside on oxygen, with U on O-2p (Section 3.4). Both solutions contain the one formal hole; the dual-U result is the one in which the spin count equals the hole count, as in the dual-U Rb cells, but it rests on a $\text{U}(\text{O-2p})$ value transferred from Cr_2O_3 and on calculations that have not been extended to the vacancy cell or to the formation-energy cycle. Until those are available, the Pr-substituted parent should be described as having one hole whose localisation depends on the treatment of oxygen correlation; it is neither closed-shell nor demonstrably trap-free. For the vacancy-containing Pr cell, the Ti-3d-only treatment places the net carrier, one excess electron, in conduction-band-derived states (Table 1 for the count, Figure 4 for the placement); this conclusion rests on a single smeared non-magnetic solution and one magnetic trial, has not been tested with the oxygen-site term, and concerns a cell with one carrier per 570 Å^3 .

4.5 Implications for dopant screening in oxide electron-transport layers. The results are consistent, within this protocol and host, with the formal charge mismatch as a candidate screening variable for how strongly a substitutional acceptor lowers the oxygen-vacancy formation energy: both dopants studied lower it, the trivalent dopant leaving the vacancy uphill at the oxygen-rich limit and the monovalent dopant making it spontaneous there. Extending this to other dopants of the same formal charge, and the three statements that follow, are hypotheses that this work motivates but does not test. (i) Trivalent rare-earth dopants other than Pr (La, Nd, Sm) should behave like Pr in the V_O energetics, because they share the -1 e mismatch. (ii) Pentavalent donors (Nb^{5+} , Ta^{5+}) should raise rather than lower $E_f(\text{V}_\text{O})$, having a $+1$ e mismatch, the electronic counterpart of the trivalent case. (iii) The acceptor states introduced by alkali substitution are candidates for recombination centres at an ETL/absorber interface. Testing (iii) requires interface models, charge-transition levels with finite-size corrections and carrier-capture calculations, none of which are reported here, so no ranking of dopants by device performance is made.

5. Limitations

- i. *Supercell and concentration.* One dopant and one vacancy per $2 \times 2 \times 1$ cell correspond to 6.25 % cation substitution and 3.1 % anion vacancies. Defect-image interactions and band filling by the net carrier of each vacancy cell are not corrected; absolute formation energies at the dilute limit may differ by several tenths of an eV, and because the Pr cell carries a conduction electron and the Rb cell a valence hole these terms need not cancel in ΔE_f . (ii) *Brillouin-zone sampling.* Production relaxations are Γ -only. Γ -only total energies lie 22–25 meV per atom below the $3 \times 3 \times 2$ values (Section 3.7); formation energies rely on cancellation within each host, and the $2 \times 2 \times 2$ check shifts each doped-host E_f by 0.10–0.14 eV and ΔE_f by 0.04 eV, which estimates rather than bounds the residual; the pristine E_f was not checked. (iii) *Plane-wave cut-off.* 40 Ry was not tested against a higher cut-off; a comparable anatase study uses 55 Ry [13]. (iv) *Force tolerance.* 0.005 Ry/Bohr (0.129 eV/Å) is loose; the Pr_VO relaxation was stopped manually and the pristine V_O relaxation ended at 0.0056 Ry/Bohr. Whether the production pristine perfect cell was relaxed has to be confirmed (Section 2.6); if it was not, the pristine formation energy and the reductions relative to it are underestimated by its relaxation energy. The effect of tighter relaxation on the energetics and on the localisation results has not been tested. **[AUTHOR ACTION: re-relax Pr_VO, Pr_perfect, Rb_VO and Rb_perfect to 0.001 Ry/Bohr and report the change in E_f and in m_{abs} .]** (v) *Spin search.* The production relaxations started from zero magnetisation (Section 2.5), and one magnetic trial was performed, for Pr_VO; occupations, smearing and symmetry settings were not varied, and two spin solutions of Rb_VO were found that differ by 6.4 meV (Section 3.3). Because E_f is a difference of total energies, a lower-energy spin solution of any parent or vacancy cell, in particular of Pr_perfect, which was not given a magnetic trial, would shift the corresponding E_f by the same amount. (vi) *Pr-4f treatment.* The Pr dataset freezes the 4f electrons in the core. Mechanisms involving 4f occupation, 4f hybridisation or a $\text{Pr}^{3+}/\text{Pr}^{4+}$ change of oxidation state cannot be examined with it, and none is claimed; the trivalent state, and with it the formal carrier counts of the Pr cells in Table 1, is imposed by this choice rather than tested. (vii) *Oxygen-site U.* $U(\text{O-2p}) = 5.5$ eV is transferred, not derived for anatase; the dual-U treatment was applied to the vacancy-free cells and Rb_VO (whose relaxation was not completed when logged) but not to Pr_VO or to the formation-energy cycles. (viii) *Charged cells.* No electrostatic or potential-alignment correction was applied; charged-cell energies are not interpreted. An ion-clamped dielectric constant of $\epsilon_\infty \approx 5.8$ (in-plane value tabulated by Arrigoni and Madsen [14]) would give a first-order monopole correction [39] of a few tenths of an eV for $q = +1$ at this cell size and four times that for $q = +2$. The static constant of anatase is much larger and would reduce the estimate, so these numbers are upper bounds and are not used. (ix) *Oxygen reference.* All formation energies are at $\mu_O = \frac{1}{2}E(\text{O}_2)$ (oxygen-rich limit) without finite-temperature corrections; equilibrium vacancy concentrations, dopant incorporation energies, the stability field of TiO_2 against lower oxides and competing phases were not evaluated, so no statement about the synthesisable phase of Rb-doped anatase is made. (x) *Defect channels and stoichiometry.* Only a single V_O per cell was computed; sequential vacancies, the fully compensated $2 \text{ Pr} + 1 V_O$ and $2 \text{ Rb} + 3 V_O$ stoichiometries, cation vacancies, interstitials and dopant clustering were not. (xi) *Functional.* The HSE06 check concerns the pristine cell only; the doped and defective cells have not been verified with a hybrid functional. (xii) *Dispersion.* The D3 correction to the formation energies

is undetermined because the parent-cell terms were not computed. (xiii) *Charge and spin analysis*. The Bader comparison is between cells that differ in composition and by two carriers. The Löwdin moments that place the Rb_VO hole mainly on one oxygen depend on the projector basis and have not been cross-checked with a spin-density isosurface or another partitioning; for Pr_perfect and the dual-U cells no site-resolved analysis exists. (xiv) *Scope*. No surface, interface, transition-level, capture-coefficient or device calculation is reported; statements about electron-transport layers are hypotheses.

6. Conclusions

Using one PBE+U protocol in the same $2\times 2\times 1$ anatase supercell, we find neutral oxygen-vacancy formation energies of +4.74 eV (pristine), +1.29 eV (Pr-substituted) and -4.06 eV (Rb-substituted) at the oxygen-rich reference. Both substitutions lower the cost of oxygen removal; Pr leaves it uphill and Rb makes it spontaneous. The Pr-Rb contrast of 5.36 eV changes by 0.04 eV between Γ -only and $2\times 2\times 2$ sampling. The ordering follows the formal electron count: one hole per Pr^{3+} and three per Rb^+ , with a single vacancy over-compensating Pr and leaving one residual hole in the Rb cell. The Rb cells carry unpaired spin in every treatment tested. In Rb_VO the residual hole is an empty minority-spin O-2p state whose Löwdin moment sits mainly on one oxygen bonded to the Rb ion (0.59 of 1.01 μ_B), and the Rb ion has moved 1.27 Å towards the vacancy, so lattice relaxation contributes to the Rb energetics; a charged-cell series reaches three unpaired spins per Rb^+ . The Pr cells converge to non-magnetic smeared solutions with U on Ti-3d only, but the vacancy-free Pr cell develops an unpaired spin when U is also placed on O-2p, so the spin state of the hole in Pr-substituted anatase is method dependent and a closed-shell, trap-free description of that material is not supported. The two-dopant comparison is consistent with charge-compensation control of oxygen-vacancy energetics in this host and provides common-protocol values for one trivalent and one monovalent dopant, but it does not separate oxidation state from ionic size and bonding, and it does not establish interfacial recombination behaviour. Completing the dual-U formation-energy cycles, tightening the relaxations and adding a matched-valence dopant series are the natural next steps.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to restructure and edit the text, to check the internal consistency of numbers, cross-references and citations, and to write the plotting scripts that draw Figures 1–4, S1 and S2 and the graphical abstract directly from the authors’ Quantum ESPRESSO outputs and computed values, without altering any data. **[AUTHOR ACTION: Figure S3 is still interim artwork whose text labels were raster-patched by figures/patch_labels.py; regenerate it from the force data of the relaxation outputs. Elsevier does not**

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CRedit authorship contribution statement

Mohammad Ghadimimehr: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Azimah Binti Omar:** Supervision, Methodology, Validation, Writing – review & editing. **Siti Rohani Binti Sheikh Raihan:** Resources, Writing – review & editing.

Data availability

The Quantum ESPRESSO input files, converged output files, pseudopotential files, Bader output tables, projected-density-of-states and Löwdin data, the full-precision energy ledger (Table S9), the run manifest (Table S10) and the Python scripts that generate every figure are deposited at Zenodo under DOI **[AUTHOR ACTION: deposit the package before submission and insert the DOI; reviewers should be able to open it during review]**.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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